

Customer Support Classification Tool

1. Objective

The objective of this task was to build an LLM-based customer-support classification tool that takes a support message as input and returns a validated JSON response containing:

- category
- priority
- sentiment
- needs_human

The output is validated using a Pydantic schema to ensure that the response follows the required structure.

2. Input

The classifier accepts a natural-language customer-support message.

Example:

I was charged twice for the same order. Please refund the extra payment.

3. Expected Output Structure

The classifier returns:

```
{  
  "category": "billing",  
  "priority": "high",  
  "sentiment": "negative",  
  "needs_human": true  
}
```

The response is parsed as JSON and validated against the defined Pydantic schema.

4. Classification Fields

Field	Description
category	Identifies the main type of customer issue
priority	Determines the urgency of the issue
sentiment	Identifies the customer's emotional tone
needs_human	Determines whether human support is required

Category Values

billing

technical

account

shipping

product

refund

other

Priority Values

low

medium

high

critical

Sentiment Values

positive

neutral

negative

Needs Human

true

false

5. Processing Workflow

Customer Support Message



Classification Prompt



Gemini LLM



JSON Response



JSON Parsing



Pydantic Validation



Validated JSON Output

If the response is invalid, the classifier attempts the request again according to the configured retry limit.

6. Technology Used

- Python — implementation language
- Google Gemini API — LLM used for classification
- google-genai — Gemini API integration
- Pydantic — JSON schema validation
- JSON — structured output format
- Pytest — automated testing

7. Testing with 20 Realistic Samples

The classifier was tested using 20 realistic customer-support messages covering different types of customer issues.

The samples included:

- Billing problems
- Account problems
- Shipping issues
- Technical problems
- Product complaints
- Refund requests
- Positive customer feedback
- Urgent requests

Sample Test Cases

#	Support Message	Main Category
1	I was charged twice for the same order.	Billing
2	I forgot my password and cannot log into my account.	Account
3	My package has not arrived for five days.	Shipping
4	The application crashes when I upload a file.	Technical
5	The new product works perfectly.	Product
6	I want to return my order and receive a refund.	Refund
7	My business account is completely locked.	Account
8	When will my order be delivered?	Shipping
9	The product I received is damaged.	Product
10	I cannot connect the application to the server.	Technical
11	I don't recognize a charge on my account.	Billing
12	My account email needs to be changed.	Account
13	My order arrived much later than expected.	Shipping
14	The website shows an error whenever I try to pay.	Technical
15	The product quality is excellent.	Product

16	Please process a refund for my cancelled order.	Refund
17	I need immediate help because my account is locked.	Account
18	My package arrived with missing items.	Shipping
19	I was billed more than the amount shown at checkout.	Billing
20	The application is not working after the latest update.	Technical

8. Test Result

The 20 samples were passed through the classifier and the resulting JSON was validated using the Pydantic schema.

The test confirmed that the classifier can:

- Accept natural-language support messages.
- Generate structured JSON.
- Produce the four required fields.
- Validate the generated response.
- Handle different customer-support scenarios.
- Reject invalid structured responses through schema validation.

Automated tests were also implemented using Pytest.

9. Example Validated Result

Input:

The application crashes every time I try to upload a file.

Validated Output:

```
{  
  "category": "technical",  
  "priority": "high",  
  "sentiment": "negative",  
  "needs_human": true
```

}

10. Conclusion

The customer-support classification tool successfully implements the required structured classification workflow.

It accepts a support message and produces a validated JSON object containing category, priority, sentiment, and needs_human.

Testing with 20 realistic samples demonstrated that the tool can be used as a reusable component for future customer-support automation systems.